



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Building Modern Web Applications.

Microsoft ASP.net

Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can say what ASP.NET is.
- I can talk about basic characteristics of the ASP.NET Web Forms Model.
- I can discuss information about components of the .Net framework.

Ask your partner:

- Which methods to create web or mobile applications do you know?
- What kind of applications would you like to create?





Let's Explore!

Albert is an expert building applications!



Instructions: Read the text below.

Elsa: Hey, did you finally create the mobile app for your business?

Albert: Yes, it was easy. I used a development platform that is called ASP.NET and I got good results.

Elsa: Cool, what exactly is ASP.NET?

Albert: It is an open source framework that offers a programming model and a software infrastructure to build robust web applications for PC and mobile devices. ASP.NET is destined to provide many programming services which make this platform very user-friendly.

Elsa: And how does it work?

Albert: ASP.NET is run with an HTTP protocol that uses different commands to establish communications between a browser and a server. The applications designed in ASP.NET are compiled codes that can use the whole hierarchy of classes available in the .NET framework. Besides that, these applications can support different programming languages as C# or JScript.

Elsa: So, we could say that the web applications built in ASP.NET are interactive and data-driven, right?

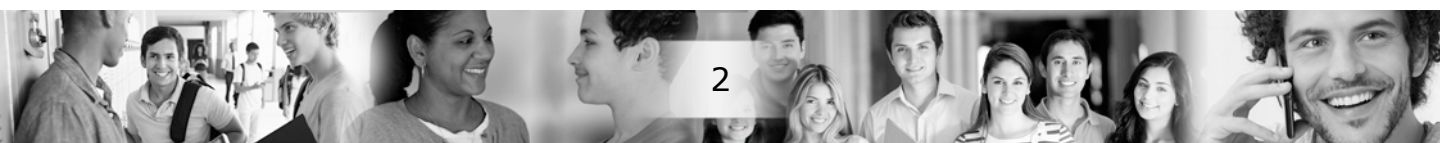
Albert: Correct. The application's interactivity is the result of the ASP.NET Web Forms Model and it has really useful tools as it is an event-driven model. For example, when the browser sends a web form to the server, the server provides a full markup page as feedback. Another feature is that ASP.NET stores information in two different ways; page state, which is the state of the client; and session state, which is a collection of information obtained from visited websites.

Elsa: This final part is not clear to me...How is this framework composed?

Albert: Well, it's important to understand that ASP.NET is based on an object-oriented model. This model contemplates every page or application as an object and all the page components are considered as objects too.

Elsa: So, what you are saying is that all server-side counterparts and server controls that are necessary when developing user-interfaces are also objects.

Albert: Exactly. I think you can help me with my next application project. You're a fast learner.





Let's Practice!

What exactly is ASP.NET?

Instructions: Read the clues. Complete the word search puzzle.

1. I used a development platform _ _ _ _ is called ASP.NET.
2. Programmers, _ _ _ use ASP.NET, can create multiple mobile apps.
3. ASP.NET is an open source framework that _ _ _ _ _ a programming model.
4. This framework provides a lot of services _ _ _ _ are very user-friendly.
5. ASP. NET is a platform that _ _ _ _ _ robust web applications for PC and mobile devices.

V	D	E	G	E	L	I	V	I	R	P	R	E	D	N	U
E	Q	N	B	O	L	S	I	T	H	A	T	H	G	H	M
L	O	F	N	O	I	T	A	T	C	E	P	X	E	S	E
J	C	R	S	W	G	C	I	L	D	T	B	A	G	C	U
V	Z	E	F	H	V	T	V	C	B	S	I	O	E	S	I
D	M	S	U	O	M	W	J	G	A	K	D	T	E	D	O
S	K	I	M	R	Y	G	R	E	K	L	G	T	V	L	T
S	E	T	E	V	A	T	E	S	O	V	L	Z	I	I	I
O	M	A	P	R	O	V	I	D	E	S	Q	Y	T	U	L
T	U	S	D	C	O	B	D	M	V	Q	O	Z	I	B	L
I	X	M	A	I	C	I	T	O	S	I	N	G	N	R	I
N	X	G	T	T	E	O	V	T	R	H	E	K	I	R	C
A	O	S	U	F	L	O	R	E	C	I	T	A	S	P	O
N	E	P	H	N	X	L	M	W	H	I	C	H	E	V	P
A	E	L	Y	U	Q	U	I	T	A	V	L	Z	D	X	E
B	M	B	U	Z	D	N	A	H	G	N	I	P	L	E	H



Let's Practice!

How does ASP.NET work?

Instructions: Read the sentences. Write the missing words in the blanks. Use: is stored in, are allowed to, is run with, is sent to, are designed.

1. ASP.NET _____ an HTTP protocol through which it is possible to establish communication between a browser and a server.
2. Web applications built in ASP.NET _____ use the whole hierarchy of classes available in the .NET framework.
3. Web applications _____ be interactive and data-driven.
4. When a web form _____ a server, the server provides a full markup page as feedback.
5. The information _____ two different ways in ASP.NET: page state and session state.



Extension Activity

Explaining the Main Features of the .NET Framework.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the .NET framework.
- Identify the main features that are offered by the .NET framework to develop dynamic web pages or applications.
- Create a slide show or a poster to explain at least 5 features of the .NET framework.
- Use examples in your explanation.
- Prepare an oral presentation to present your slide show or poster.
- Share your work with the class.

Self-Assessment

- I look for more information about the .NET framework.
- I identified the main features that are offered by the .NET framework to develop dynamic web pages or applications.
- I explain at least 5 features of the .NET framework.
- I share my work with the class.

Yes No

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Target Words

- **feedback** (noun): response to an output that sometimes checks if it needs to be corrected
e.g. When the browser sends a web form to the server, the server provides a full markup page as feedback.
- **interactivity** (noun): the range in which components are able to create a contact among themselves
e.g. The application's interactivity is the result of the ASP.NET Web Forms Model.
- **robust** (adjective): with a strong shape
e.g. It is an open source framework that offers a programming model and a software infrastructure to build robust web applications for PC and mobile devices.
- **protocol** (noun): a series of actions organized by agreement
e.g. ASP.NET is run with an HTTP protocol that uses different commands to establish communications between a browser and a server.
- **hierarchy** (noun): a system that measures its parts by importance
e.g. The applications designed in ASP.NET are compiled codes that can use the whole hierarchy of classes available in the .NET framework.
- **platform** (noun): a program where applications can be designed
e.g. I used a development platform that is called ASP.NET and I got good results.



Learning Points

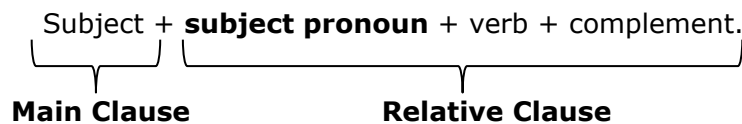
Relative Clauses: Subject Pronouns.

You use relative clauses to provide extra information about a subject or an object mentioned in a previous clause.

To give additional information about a subject, we need to use subject pronouns such as **who**, **which**, and **that**.

When the relative pronoun is used as a **subject pronoun**, the main verb goes after it. Let's see the grammatical structures below:

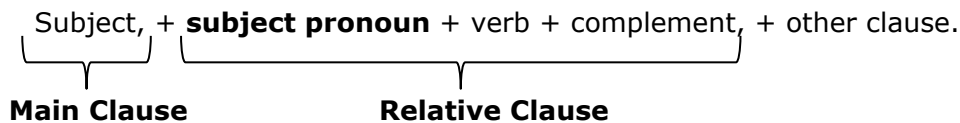
Defining Relative Clauses



Examples:

- *This is a manual **that** covers everything about ASP.NET*
- *I read a manual **which** explains how to use ASP.NET*
- *She is the instructor **who** recommended me to read that manual.*

Non-Defining Relative Clauses



Examples:

- *The manual, which was really interesting, explained all the ASP.NET Component Model.*
- *The professor, who is a really good instructor, recommended us a very useful manual.*

Examples

- *I used a development platform that is called ASP.NET.*
- *It's an open source framework that offers a programming model and a software infrastructure to build robust web applications for PC and mobile devices.*
- *Another feature is that ASP.NET stores information in two different ways: page state, which is the state of the client; and session state, which is a collection of information obtained from visited websites.*





Learning Points

Components of .NET Framework 3.5.

You use **.NET Framework 3.5** as a Microsoft code platform that provides tools and libraries for the development of applications.

.NET Framework includes a series of components that perform a specific task in the development and management process.

The following list shows some of the main components and the task they perform:

1. **Common Language Runtime (CLR):** It is in charge of memory management. It handles exceptions, execution of threads and codes. It also validates all the security protocols.
2. **.NET Framework Class Library:** It provides reusable types of interfaces, structures, and values.
3. **Common Language Specification:** It defines supported languages and their integration in the framework.
4. **Common Type System:** It offers a guideline to manage different types when running.
5. **Metadata and Assemblies:** It contains the binary information and gathering resources.
6. **Windows Forms:** It graphically represents windows displayed in an application.
7. **ASP.NET and ASP.NET AJAX:** allows the user to update data on a website without having to disassemble it completely. ASP. NET is the development model and AJAX allows to update data on a page.
8. **ADO.NET:** It provides access to data sources to retrieve, manipulate, and update data.
9. **Windows Workflow Foundation (WF):** It creates work-flow based applications for Windows.
10. **Windows Presentation Foundation:** It develops visual interfaces by using documents, graphics, media, and other items.
11. **Windows Communication Foundation (WCF):** It is in charge of creating and executing connected systems.
12. **Windows CardSpace:** It provides the appropriate security when a user has access to or shares information online.
13. **LINQ:** It controls querying capabilities to .NET languages.